VISHVKISAN

TECHNICAL DESIGN DOCUMENT

System Architecture • Modules • Data Flow • Security • PPP Model

From Fields to Families

A Digital Bridge for Indian Agriculture

Project: VishvKisan — digital agricultural marketplace prototype

Team: Team DigiKisan

Theme: Rural Development & Livelihoods

GitHub: https://github.com/dhitishmankumar-ctrl/VishvKisan-final-prototype-

Prototype: https://vishv-kisan-showcase--dhritishmankuma.replit.app

Status: This document separates the current demonstration prototype from planned integrations and proposed real-world operations. It does not certify that every feature, security control, partnership, budget or outcome is live or approved.

1. Project Overview

VishvKisan is a web-based agricultural marketplace prototype intended to connect farmers with customers and bring selected agricultural information and tools into one digital experience. It focuses on market access, price awareness, farming information, government-scheme awareness, community learning and responsible pesticide-use education.

1.1 Problem statement

- **Farmer income and market access:** intermediaries and limited direct-buyer access can reduce the share farmers receive; farmers may sell at low prices while consumers pay substantially more.
- **Price transparency:** farmers may lack timely, comparable market-price information for informed selling decisions.
- **Government schemes:** information about schemes, eligibility, subsidies and applications can be fragmented or difficult to access.
- **Post-harvest losses:** gaps in collection, storage, cold-chain access, packaging and transport can damage produce or limit market reach.
- **Digital and knowledge barriers:** language, digital literacy and fragmented information can make online services difficult to use.
- **Consumer transparency and safety:** buyers may want clearer information about produce origin, quality and responsible pesticide practices.

1.2 Objectives

- Provide a digital place to showcase produce and explore customer demand.
- Make displayed prices and simple profit estimates easier to compare.
- Bring crop, weather, farming, scheme and pesticide-safety information together.
- Design for rural users and future regional-language accessibility.
- Define a pathway from prototype to a measured Assam pilot.

1.3 Scope and limitations

The current deliverable is a demonstration prototype. Production-grade authentication, persistent database, live market/weather feeds, payments, official farmer verification, fulfilment logistics and government/private partnerships must be confirmed, implemented and tested before being described as operational.

2. Technology Stack and Prototype Modules

Technology / area	Role	Implementation note
HTML	Web structure	Provides structure for interface pages.
CSS	Styling / layout	Visual presentation and responsive styling where implemented.
JavaScript	Client interactions	Supports interactive behaviour where implemented.
TypeScript	Typed application code	Present in repository; exact usage should be checked module by module.
pnpm workspace	Package management	Consult workspace files and manifests for exact setup commands.
GitHub / Replit	Source control and development/deployment workflow	Supplied repository and hosted prototype links.

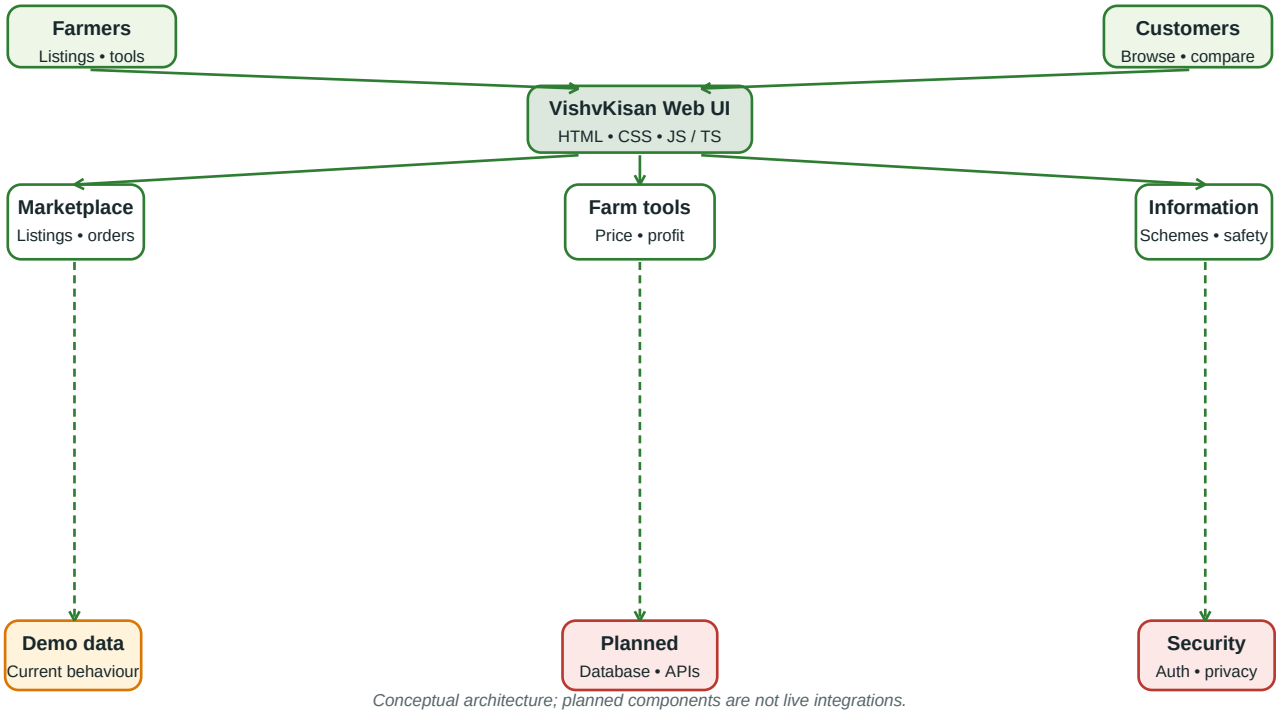
2.1 Main modules

Module	Purpose	Status caveat
Farmer workspace	Farmer tools, listing management and profit calculator where implemented.	Test each action in the live prototype.
Customer marketplace	Browse produce, inspect details, search/compare displayed prices and explore order flows.	Listings may be demo data; verify search and order behaviour.
Price / profit tools	Compare displayed prices and estimate selling outcomes.	Do not call external prices live unless verified.
Crop / weather	Crop information, farming tips and weather-related guidance.	Live feeds need integration and validation.
Government schemes	Scheme awareness and information for further checking.	Verify eligibility and current status with official sources.
Farmer licence / e-card	Concept for farmer details and a possible verification workflow.	A prototype badge is not official verification.
Pesticide safety	Responsible-use awareness and safer farming practices.	No residue-free claims without appropriate testing.
Community / orders	Knowledge-sharing concepts and demonstration order flows.	Real fulfilment requires backend and operations.

Repository hygiene: verify setup commands from the actual manifests, keep API keys/passwords out of GitHub, and document environment-variable names without publishing secret values.

3. System Architecture

This conceptual diagram separates the user-facing prototype from data, integrations and security capabilities that may need further implementation. It is not proof that every layer is deployed.



Conceptual architecture; planned components are not live integrations.

3.1 Architecture layers

Layer	Responsibility	Verification required
Users	Farmers and customers; admin roles are proposed if needed.	Confirm actual account roles and controls.
Web interface	Marketplace, farmer tools, information modules and navigation.	Test responsive layout, accessibility and navigation.
Application modules	Listings, price display, profit calculation, information and demo order flows.	Test validation, edge cases and calculations.
Data layer	Potential user, listing, order and information records.	Confirm local/demo versus persistent storage.
External services	Potential weather, market, government info, maps and payment providers.	Mark planned unless integrated and tested.
Security	Authentication, access control, privacy, audit and verification procedures.	Production controls require review.

3.2 Future production architecture

A production version would separate the web client, API/application services, persistent database, third-party integrations and operational administration. Access should be role-based, sensitive data minimised, and integration failures handled safely.

4. User Journeys and Data Flow

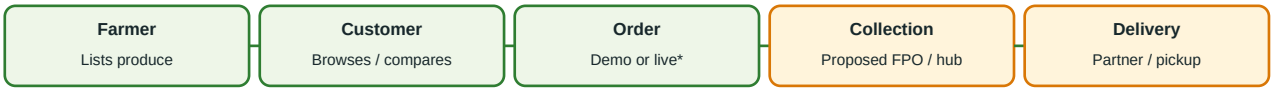
4.1 Farmer journey

- Open the platform and choose the farmer workspace.
- Explore tools and review displayed information.
- Create or update a produce listing where implemented.
- Review price information and use the profit calculator if available.
- Manage listing/order details and provide feedback according to current prototype behaviour.

4.2 Customer journey

- Open the customer marketplace.
- Browse/search produce and review product information.
- Compare displayed prices and product details.
- Use the available order action; call it a demonstration flow if no live order backend exists.
- Provide feedback on usability, trust and product-information clarity.

4.3 Proposed farm-to-customer logistics



**Use "demonstration" unless a real order backend is implemented and tested.*

Proposed sequence: farmer/FPO supply → collection and aggregation → quality check → storage/cold chain where needed → sorting and packing → local delivery partner or pickup → customer. Each stage requires partners, service standards, cost estimates and responsibility allocation.

4.4 Data-flow considerations

- Product listings should include only necessary information.
- External price comparisons should identify source, unit, location and date.
- Order records must distinguish simulated orders from confirmed real transactions.
- Weather, scheme and crop information should include source and freshness checks where possible.
- Administrative changes to verified records should be auditable in a production system.

5. Security, Privacy and Food Safety

Area	Recommended control	Status
Authentication	Use tested authentication; protect account recovery and apply appropriate roles.	Implement/test before production.
Data minimisation	Collect only needed information and explain why it is collected.	Policy and implementation required.
Transport/storage	Use secure transport, restrict database access and protect stored personal data.	Validate before real user data.
Payments	Use compliant payment provider; never store payment credentials directly.	Planned unless tested.
Farmer verification	Define authorised verifier, evidence, renewal, corrections and dispute handling.	Proposed; requires partner.
Audit/reporting	Log administrative actions and provide reporting for inaccurate listings.	Process to define.

5.1 Responsible pesticide-use information

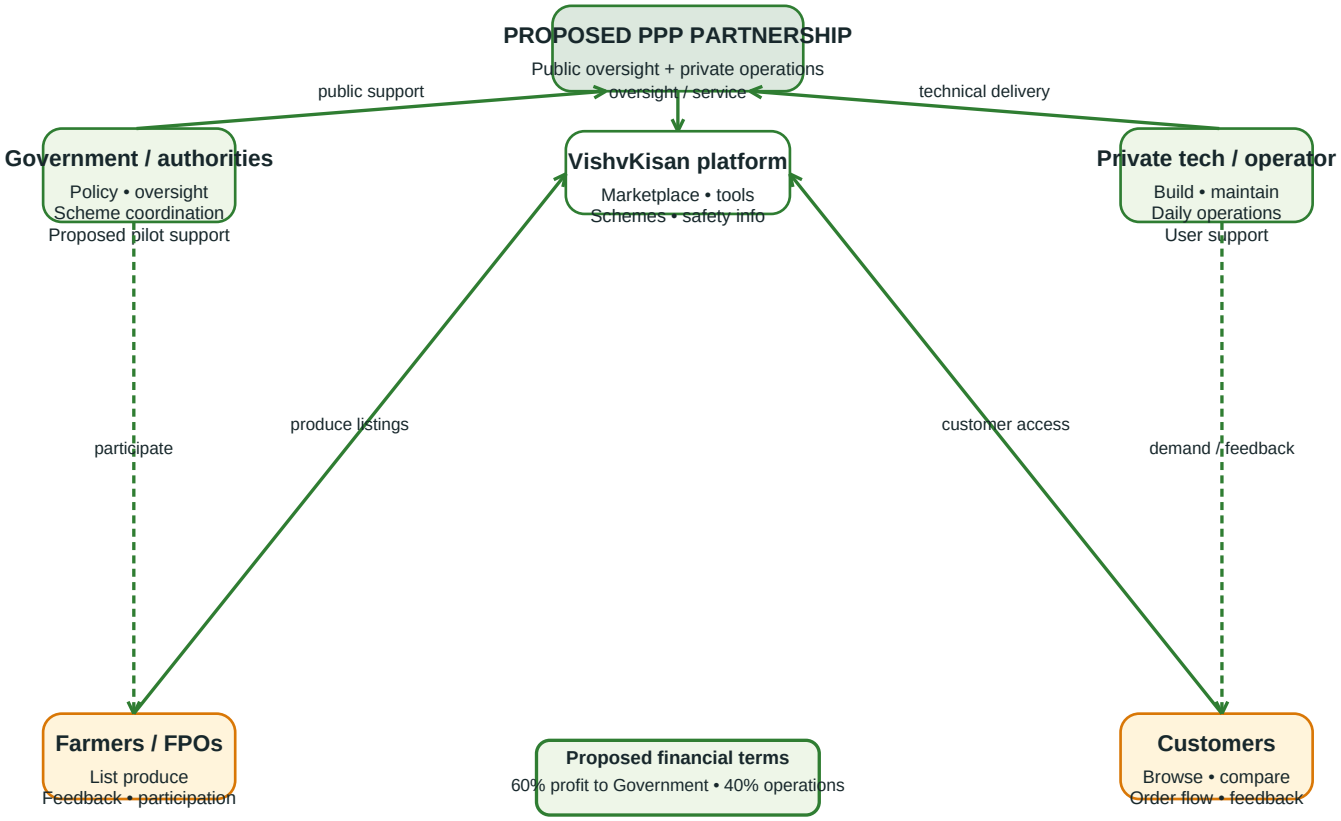
The platform can share awareness about label directions, appropriate use, training, integrated pest management concepts and harvest waiting periods using authoritative agricultural guidance. Farmer identity verification alone does not prove produce is pesticide-free or residue-free; such claims require defined testing and verification.

5.2 Testing checklist

- Confirm pages load and navigation works.
- Test desktop/mobile layouts, listings, search, filters, price displays and product details.
- Test forms, validation, empty states, error messages and calculations.
- Verify external links and data freshness where integrations exist.
- Clearly label demo orders and sample prices.
- Test access control and privacy before collecting real user data.
- Validate pesticide-safety content against authoritative agricultural guidance.

6. Proposed Public–Private Partnership (PPP) Model

This is a proposed operating arrangement, not evidence of an approved government agreement. It requires partner selection, legal review, funding approval, procurement/compliance review and a signed agreement.



Proposal only; requires approval, legal review and a signed agreement.

6.1 Proposed roles

Participant	Proposed responsibilities
Government of Assam / authorities	Policy/institutional support; scheme coordination; oversight; possible authorised verification support; proposed pilot funding subject to approval.
Private technical / operating partner	Platform development and maintenance; technical integrations; daily operations; user support and improvements.
Farmers / FPOs	List produce, participate, provide feedback and follow applicable safety/trading requirements.
Consumers	Browse produce, compare information, use order flows and provide feedback.

6.2 Proposed financial assumptions

Item	Proposal
Pilot geography	Two to three Assam districts; expansion only after evaluation.
Indicative funding	■1–2 crore proposed; not an approved allocation or confirmed budget.
Profit / operations split	Proposal states 60% of profit to government and 40% for operational costs—not 60% to the private partner.
Contract conditions	Define revenue, expenses, net profit, audit rules, timing, losses, ownership, data governance and termination terms through legal review.

6.3 Main risks and mitigations

Risk	Proposed mitigation
Slow adoption / low trust	Village demonstrations, simple/local-language interface, field agents and clear verification.
Fund misuse / poor governance	Audits, quarterly reporting, milestone-linked funding and transparent procurement.
Overlap with ONDC, e-NAM or mandis	Position as complementary where appropriate; assess interoperability later.
Storage/transport gaps	Start in selected districts using existing FPOs, collection centres and local transport.
Low revenue	Conservative assumptions, staged investment and single-district fallback.
Privacy / security	Minimise data, restrict access and review applicable data-protection duties.

Risk	Proposed mitigation
Seasonal volatility	Diversify produce and provide clearly sourced price information; avoid fixed-price promises.

7. Roadmap, KPIs and Evaluation

Phase	Focus	Exit criteria
1. Prototype refinement	Improve interfaces, document functions, test journeys and limitations.	Core pages/demo flows reviewed; setup documented.
2. Technical foundations	Add/test authentication, persistent data, roles and integrations.	Security/integration checks pass before real data use.
3. Pilot preparation	Engage partners; define verification, logistics, training, support, budget and governance.	Agreements, funding and operating plan approved.
4. Pilot and evaluate	Run limited pilot, gather feedback, audit outcomes and improve.	Review results against baseline before expansion.

7.1 Suggested KPIs

KPI	Measurement
Farmer participation	Registered/active farmers; verified records only after authorised process exists.
Customer adoption	Registered/active customers, repeat use and feedback.
Marketplace activity	Listings, enquiries, orders and completion rate; distinguish demo from real orders.
Livelihood outcomes	Compare farmer net earnings and selling costs with a documented baseline.
Food-safety awareness	Training participation and adoption of recommended practices; residue claims require testing.
Service reliability	Page performance, errors, successful workflows and support response times.

7.2 Intended impact

VishvKisan aims to improve market access, agricultural information access, scheme awareness, responsible farming and consumer transparency. Increased income, lower consumer prices, reduced losses and food-safety improvement are hypotheses to measure during a properly designed pilot, not results already achieved.

8. Repository Documentation Plan

- **README.md:** overview, current status, screenshots, tested setup/run commands and limitations.
- **TECHNICAL_DESIGN.md / PDF:** architecture, modules, user journeys, data flow, security, PPP and roadmap.
- **Testing checklist:** repeatable manual or automated tests.
- **LICENSE:** choose a software licence after deciding reuse terms.
- **Configuration:** document setup and environment-variable names only after verifying repository files; never commit secrets.

9. Project Links and References

GitHub: https://github.com/dhitishmankumar-ctrl/VishvKisan-final-prototype-

Demonstration prototype: https://vishv-kisan-showcase--dhritishmankuma.replit.app

Use the project proposal as the source for proposed pilot scope, PPP assumptions, risks and intended impact. Confirm scheme information and pesticide guidance with authoritative sources. Confirm implementation claims from the repository and live tests.

Final status: This document describes the supplied project concept and technical/proposal information. It does not certify that all modules, integrations, security controls, partnerships, funding arrangements or outcomes are operational or approved.